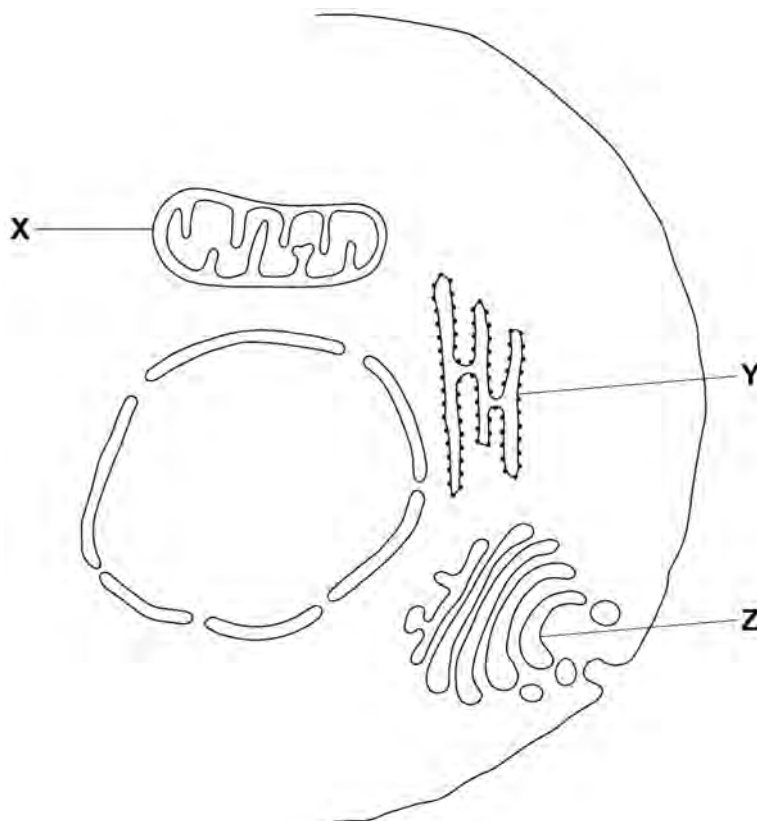


0	2
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Figure 3 shows part of a eukaryotic cell.

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outside the
box

Figure 3



0	2	.	1
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Name the organelle labelled X and state its function.

[2 marks]

Name _____

Function _____



0	2	.	2
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Pancreatic cells produce and secrete digestive enzymes.

Explain why pancreatic cells contain large numbers of organelles **Y** and **Z**.

[4 marks]

Question 2 continues on the next page

Turn over ►

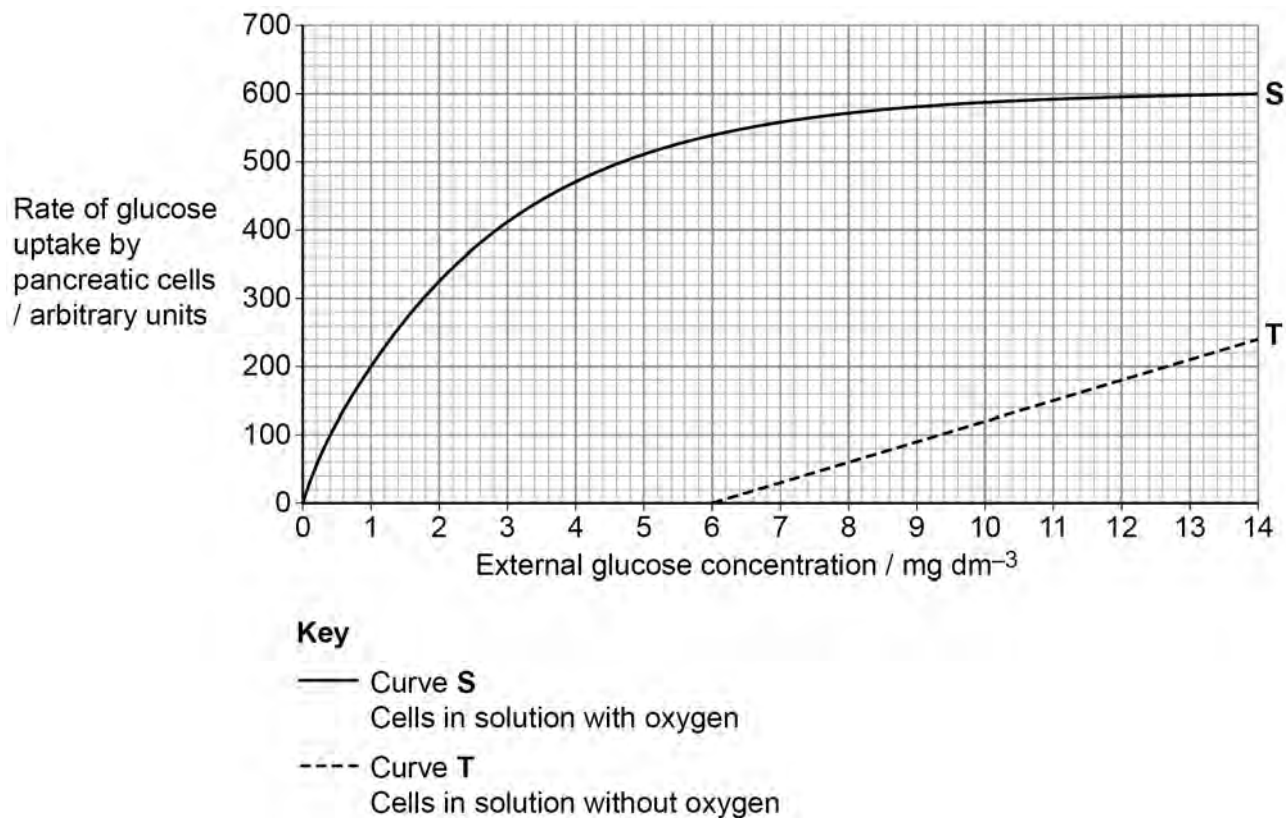


0 2 . 3

A scientist placed pancreatic cells in solutions containing different concentrations of glucose. The scientist measured the rate of uptake of glucose by these cells.

Figure 4 shows the scientist's results.

Figure 4



Using **Figure 4**, what can you conclude about the processes involved in the uptake of glucose molecules by pancreatic cells?

[3 marks]



0 2 . 4

Use **Figure 4** to calculate the ratio

rate of glucose uptake with oxygen : rate of glucose uptake with no oxygen
at an external glucose concentration of 14 mg dm^{-3}

[1 mark]

Ratio = _____

10**Turn over for the next question****Turn over ►***Do not write
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